Ch 7.4: Cubic splines

Lecture 23 - CMSE 381

Michigan State University
::
Dept of Computational Mathematics, Science & Engineering

Mon, Mar 23, 2026

Announcements

Last time:

- 7.2 Step functions
- 7.3 Basis functions

This lecture:

- 7.4 Cubic splines

Announcements:

- Homework # 5 is due Sunday.

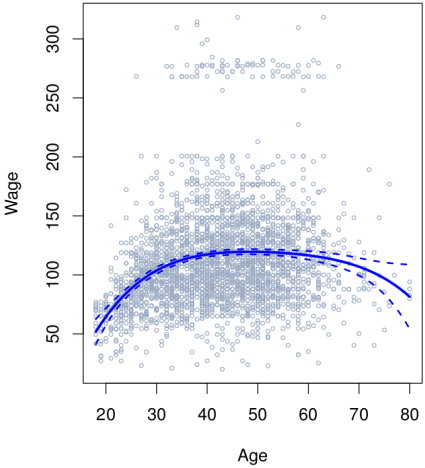
21	W	3/18	Polynomial & Step Functions	7.1-7.2		
22	F	3/20	Step Functions; Basis functions; Start Splines	7.2-7.4		
23	M	3/23	Regression Splines	7.4		
24	W	3/25	Decision Trees	8.1		Q7
25	F	3/27	Random Forests	8.2.1, 8.2.2	HW #5 Due Sun 3/29	
26	M	3/30	Maximal Margin Classifier	9.1		
27	W	4/1	SVC	9.2		Q8
28	F	4/3	SVM	9.3, 9.4		
29	M	4/6	Single Layer NN	10.1		
30	W	4/8	Multi Layer NN	10.2		Q9
31	F	4/10	CNN	10.3	HW #6 Due Sun 4/12	
32	M	4/13	Unsupervised learning / clustering	12.1, 12.4		
33	W	4/15	Virtual: Project Office Hours			Q10
	F	4/17	***Review***			
	M	4/20	**Midterm #3**			
	W	4/22				
	F	4/24			**Project Due**	

Section 1

Previously

Polynomial regression

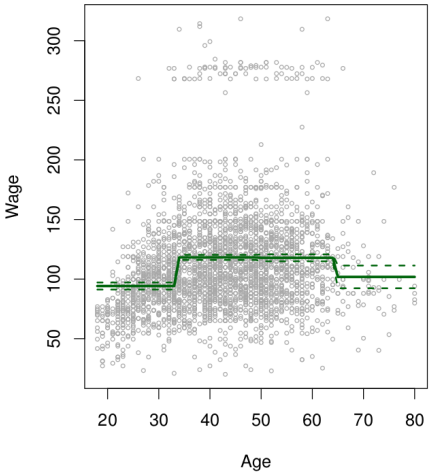
$$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \varepsilon_i$$



Step function regression

$$
\begin{aligned}
C_0(X) &= I(X < c_1), \\
C_1(X) &= I(c_1 \le X < c_2), \\
C_2(X) &= I(c_2 \le X < c_3), \\
&\vdots \\
C_{K-1}(X) &= I(c_{K-1} \le X < c_K), \\
C_K(X) &= I(c_K \le X),
\end{aligned}
$$

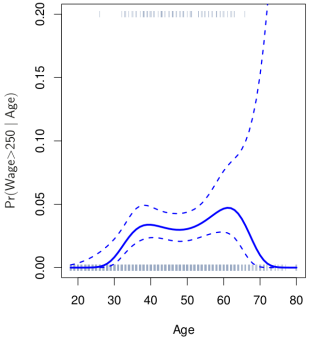
$$y_i = \beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \cdots + \beta_K C_K(x_i) + \varepsilon_i$$



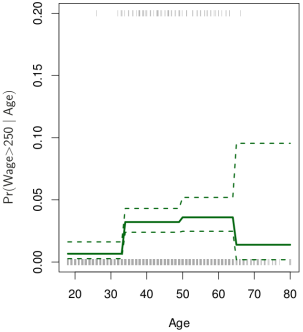
Classification version

$$\Pr(y_i > 250 \mid x_i) =$$

$$\frac{\exp(\beta_0 + \beta_1 x_i + \cdots + \beta_d x_i^d)}{1 + \exp(\beta_0 + \beta_1 x_i + \cdots + \beta_d x_i^d)}$$



$$\frac{\exp(\beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \cdots + \beta_K C_K(x_i))}{1 + \exp(\beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \cdots + \beta_K C_K(x_i))}$$



Basis Functions Setup

Polynomial and piecewise-constant regression models are special cases of a *basis function* approach.

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \cdots + \beta_K b_K(x_i) + \varepsilon_i$$

- Pick a collection of basis functions $b_1(X), \cdots, b_K(X)$
- Use least squares to figure out the constants

- Lots of possible options for these
  - Some examples are wavelets or Fourier series
  - Next section is we'll talk about regression splines

What will you learn today?

- How to define cubic splines mathematically in terms of piecewise polynomials? (review)
  - Write the equations for the model on each interval between knots.
  - Write the continuity constraints (function, first and second derivatives) at each knot.
  - Compute the total degrees of freedom.
- How to define cubic splines in terms of basis functions?
  - Describe them mathematically using the truncated power basis (textbook version).
  - Visualize them as B-spline basis functions (e.g., in Python / scikit-learn).
- How to fit a cubic spline model to data in Python?
- How to change the model flexibility?
  - What are the relevant metaparameters?
  - How to choose appropriate flexibility?
- What precautions must you take at the outer boundary? Why?

Section 2

Regression Splines

Piecewise polynomials

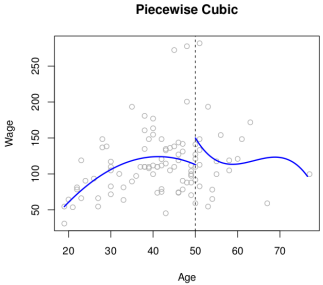
- Fit a polynomial regression

$$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \varepsilon_i$$

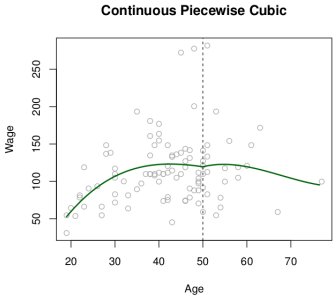
- Let the β_i's be different at different locations of the range.

- Points where the coefficients change are called "knots"
- A piecewise cubic with no knots is just standard cubic polynomial
- More knots gives more flexible polynomial
- Piecewise constant of previous section is degree 0 version of this

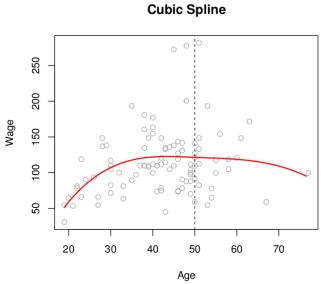
Building up to cubic splines



Piecewise cubic but discontinuous



Continuous but pointy join



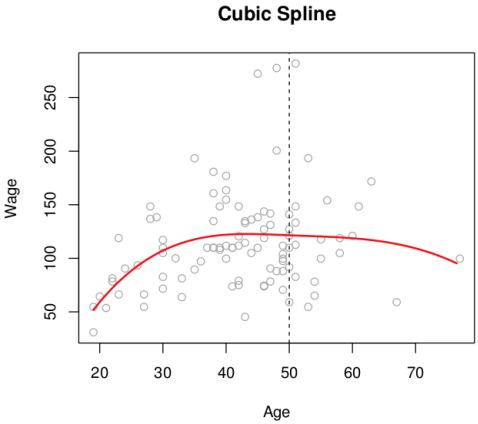
Smooth join. Same value plus same first and second derivatives.

$$y_i = \begin{cases} \beta_{01} + \beta_{11}x_i + \beta_{21}x_i^2 + \beta_{31}x_i^3 + \epsilon_i & \text{if } x_i < c \\ \beta_{02} + \beta_{12}x_i + \beta_{22}x_i^2 + \beta_{32}x_i^3 + \epsilon_i & \text{if } x_i \geq c. \end{cases}$$

Cubic splines: degrees of freedom

$$f(x) = \begin{cases} \beta_0^1 + \beta_1^1 x + \beta_2^1 x^2 + \beta_3^1 x^3 & x < c \\ \beta_0^2 + \beta_1^2 x + \beta_2^2 x^2 + \beta_3^2 x^3 & x > c \end{cases}$$

- 8 original degrees of freedom, plus three constraints (continuous, 1st deriv continuous, 2nd deriv continuous) leaves 5 degrees of freedom
- cubic spline with K knots uses $4 + K$ degrees of freedom



Spline basis representation

Want to pick b_i so that we represent a cubic spline with K knots as

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \cdots + \beta_{K+3} b_{K+3}(x_i) + \varepsilon_i$$

Warning: There is no unique way to do this. The book has a different way. We're going to use scikit learn's way.

Version 1: Truncated power basis function

$$h(x,z) = (x-z)_+^3 = \begin{cases} (x-z)^3 & \text{if } x > z \\ 0 & \textit{else} \end{cases}$$

Desmos link: https://www.desmos.com/calculator/ahllu5glar

- Book uses $z = \xi$ notation, but i refuse
- Draw $h(x, 2)$

The (book) basis for cubic splines

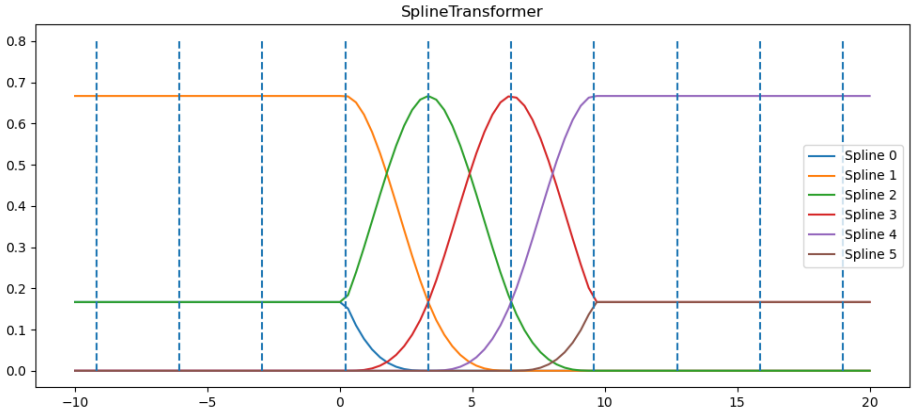
Given knots at $z_1, \cdots, z_K$

- X
- X^2
- X^3
- $h(X, z_1)$
- $h(X, z_2)$
- $\vdots$
- $h(X, z_K)$

- Adding $h(X, z_i)$ leads to discontinuity at most at the third derivative at z_i
- So use the $3 + K$ functions listed here, result is $K + 4$ coefficients once you get the intercept.
- Matches with our $K + 4$ degrees of freedom from earlier

$$f(X) = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \beta_4 h(X, z_1) + \beta_5 h(X, z_2) + \cdots + \beta_{k+3} h(X, z_K)$$

Version 2: B-spline basis function



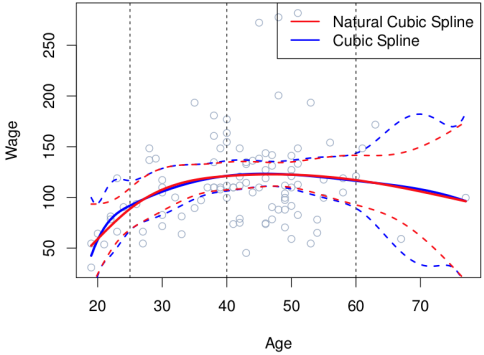
Similar idea. Take some functions that are all continuous and differentiable everywhere and add them up so you end up with a function that is continuous and differentiable everywhere.

Test your understanding: PollEv

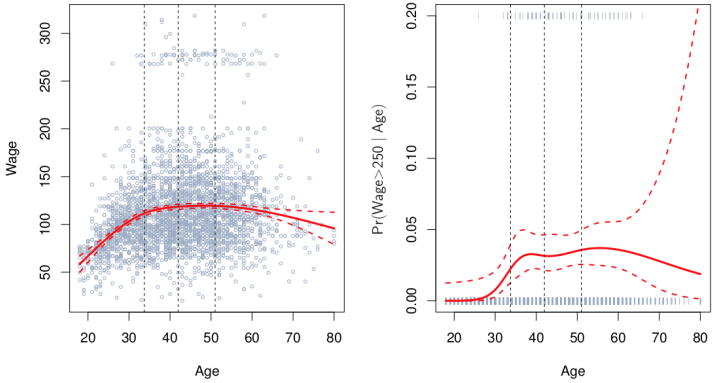
Coding example

Notes on cubic splines

- Human eyes can't detect the discontinuities so often cubic splines are a good way to go
- Still have problem of high variance at the outer range of predictors (big or small X)
- *Natural spline:* regression spline plus boundary constraints where function is required to be linear at the boundary



Where to put the knots?

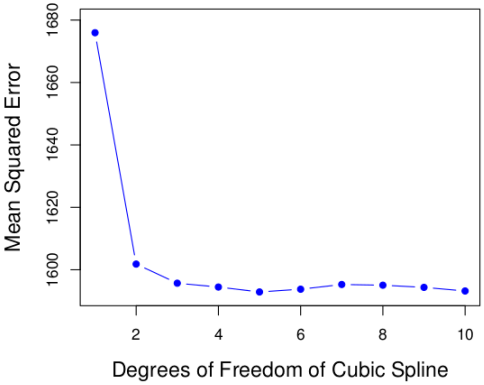


- Plan A: Uniform placement
- Plan B: Data driven. Example here puts it at quantiles of the data

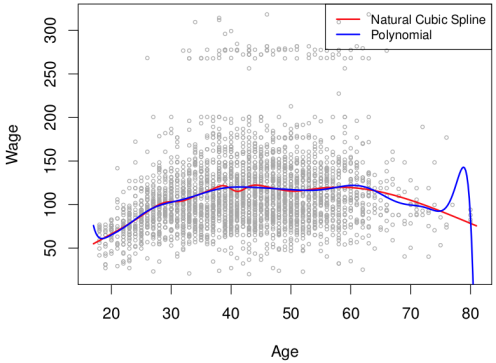
How many knots to use?

When in doubt, Cross-Validate.

- Do CV (k-fold probably) on data for each number of knots
- Figure at right is the MSE resulting from 10-fold CV
- Note that clearly linear is not enough, after that 3-4 degrees of freedom for the cubic spline are enough



Cubic splines vs Polynomial Regression



- Left: natural cubic spline with 15 degrees of freedom vs degree 15 polynomial
- Polynomial flexibility means ugly boundary behavior
- Splines let you add flexibility as more knots rather than more degrees, so allows for flexibility where the function is changing more rapidly

Next time

21	W	3/18	Polynomial & Step Functions	7.1-7.2		
22	F	3/20	Step Functions; Basis functions; Start Splines	7.2-7.4		
23	M	3/23	Regression Splines	7.4		
24	W	3/25	Decision Trees	8.1		Q7
25	F	3/27	Random Forests	8.2.1, 8.2.2	HW #5 Due Sun 3/29	
26	M	3/30	Maximal Margin Classifier	9.1		
27	W	4/1	SVC	9.2		Q8
28	F	4/3	SVM	9.3, 9.4		
29	M	4/6	Single Layer NN	10.1		
30	W	4/8	Multi Layer NN	10.2		Q9
31	F	4/10	CNN	10.3	HW #6 Due Sun 4/12	
32	M	4/13	Unsupervised learning / clustering	12.1, 12.4		
33	W	4/15	Virtual: Project Office Hours			Q10
	F	4/17	***Review***			
	M	4/20	**Midterm #3**			
	W	4/22				
	F	4/24			**Project Due**	